

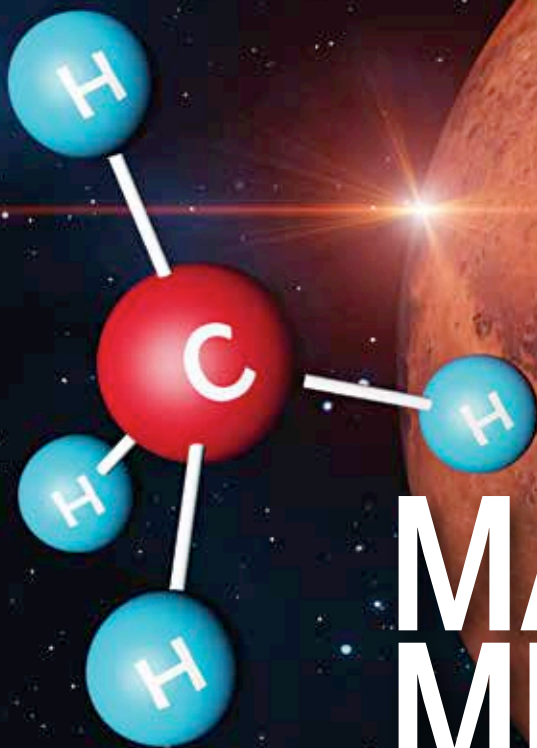
Matching spikes

For the first time, a methane spike was recorded by two probes in 2019, Curiosity and Mars Express. <https://digit.in/jan20-13>



Methane chemistry

Aarhus researchers show how Martian minerals could play a role in the vanishing methane. <https://digit.in/jan20-14>



MARS METHANE MYSTERY

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ethane was first detected on Mars by ESA's Mars Express, in 2004. The Planetary Fourier

Spectrometer (PFS) on board used spectroscopy to determine that the methane levels on Mars were 10 parts per billion by volume (ppbv). This means that for every billion molecules in the atmosphere of Mars, only 10 were methane. By comparison, Earth has a methane concentration of 1800 ppbv.

NASA's Curiosity rover detected a background level of 0.3 to 0.7 ppbv in 2012. In 2013 and 2019 Curiosity measured two spikes in methane over the background levels. The latter spike is the highest ever measured, with a concentration of 21 ppbv. ESA's Trace Gas Orbiter (TGO) also detected a spike a day after the Curiosity detection in 2019. However, Mars Express did not detect a spike in methane levels after the 2013 measurement by Curiosity. Earth based telescopes and other instru-

Orbiters and rovers have detected an ebb and flow of methane on Mars, but scientists do not know how it is produced, or where it disappears to.

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ments seemed to be reporting widely different levels of methane on Mars. After scientific doubt on the validity of the detections, and subsequent observations, it became clear that the concentration of methane in the Martian atmosphere is dynamic.

Methane is a biosignature, one of those chemicals that are produced by lifeforms. Even if the source of methane is biotic, it does not mean that those Martians are on Mars right now. The methane produced by ancient Martians might be trapped below the surface, which are now periodically being released. These traps could be made of solid forms of water, and are called clathrate

hydrates. The region on Mars that Curiosity is studying, known as the Gale crater has been found to be an ancient lake bed, suitable for life in the distant past. In fact, all the regions where methane has been observed on Mars, Terra Sabae, Nili Fossae and Syrtis Major are associated with indications of ancient rivers. The source of the methane could be the decomposition of materials containing it. Methane is also produced through abiotic processes, in natural reactions between hot water and rocks, volcanic activity or geothermal vents. 95 percent of the methane in Earth's atmosphere is produced by living things.